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RESEARCH ARTICLE

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Assessment of nationwide emergency systems in Thailand, a middle-income country setting with UHC

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Summary

Background: Evidence has been limited regarding broader emergency systems assessment in low- and middle-income countries. The aim of the present study was to provide the empirical evidence of Thai emergency department (ED) workforce vis-à-vis workload on a national scale, the availability of services for selected high-priority health burdens, and the governance obstacles in addressing the workforce management.

Method: One hundred thirty public Thai EDs that provide 24-hour emergency medical care were identified across Thailand as meeting the inclusion criteria. The mailed questionnaires were administered to collect data related to the objectives of the research study.

Result: Responses were received from 91 of 130 (70%) hospitals. The median number of patients visiting hospital EDs was 51 221 per year with 32.8% considered nonurgent (ESI levels 4-5). University hospital EDs were staffed with a higher number of ER professionals than EDs of service-based secondary care and tertiary care hospitals under Ministry of Public Health (MOPH). Almost all hospitals expressed concerns about the deficiency of doctors and nurses especially emergency physician (EP) and emergency nurses. The percentage of hospitals reporting the availability of coronary artery catheterization (34%) and thrombolytic infusion for acute thrombotic stroke (24%) was limited. The governance obstacles to manage emergency systems were considered.

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Conclusion: ED staffing seems to positively correlate with workloads except university hospitals, in our study, which may suggest the influence of teaching status on the allocation of the human resource. Among the governance obstacles in ED management, a better response to nonurgent patients requires flexibility for hospitals to set financial disincentives or mobilization and management of human resources.

KEYWORDS

assessment, emergency department, middle-income country, national health coverage, systems

1 | INTRODUCTION

In an effort to promote emergency care, the World Health Assembly passed resolution 60.22 in 2007 requesting that the director-general provides support to member states for design of quality improvement programs and other methods needed for competent and timely provision of essential trauma and emergency care.[1] In addition, improving emergency care is key to meeting these high-level commitments to the Sustainable Development Goals (SDGs) and universal health coverage (UHC).[2]

For governments to take corresponding actions, systems assessment is an area and approach to contextualize existing knowledge. The systems assessment may include essential emergency care functions, health workforce management, financial arrangement, and governance functions.[3] UHC has been embraced by World Health Organization (WHO) as a means to improve health and reduce the financial burden from receiving care. However, recent evidence has shown that among 137 low- and middle-income countries (LMICs), poor-quality health care contributed to more deaths compared with those who did not use care for each illness.[4] Evidence has been limited regarding broader emergency systems assessment in LMICs despite the resolution and the commitments.

Using mixed methods, Japiong et al[5] reported from Ghana concerning marked deficiencies in many essential items and services such as basic airway supplies, chest tubes, and several emergency medications. Similarly, substantial deficiencies in capacity were demonstrated across a range of indicators and predominantly affecting publicly funded facilities in Sierra Leone.[6] Capacity was the weakest in the domain of infrastructure, with an average score of 43% using a structured set of minimum standards necessary to deliver emergency and critical care.[6]

To address a deficiency in the health workforce, the creation of a new nonphysician clinician cadre or the emergency care practitioner provides an example of how emergency care can be practically implemented in Uganda, in a low-resource setting in which physician numbers are limited.[7] In Rwanda, Aluisio et al[8] demonstrated a postgraduate diploma program in emergency medicine (EM) resulted in a 43% reduction in overall hospital mortality.

We believe that this present report is the first evidence that shows broader emergency systems assessment in middle-income country settings with UHC. We focused on a national scale of workforce vis-à-vis workload and availability of services for selected high-priority health burdens, as well as governance obstacles in addressing the workforce management and the need for service innovations in Thailand, a middle-income country with over a decade of UHC.

2 | METHODS

This study was approved by the Institutional Ethical Review Board of the Faculty of Medicine, Ramathibodi Hospital, Thailand. Since the participation in the survey was voluntary, consent was assumed by the voluntary choice of participating from July to September 2018.

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2.1 | Study setting

In 2014, 67% of the country's 161 000 Thai hospital beds were in the Ministry of Public Health (MOPH) facilities, 14% of beds were in other public, non-MOPH facilities, and 19% of beds were in private hospitals.[9] In regard to emergency and trauma medical care, MOPH hospitals are organized into a multitier system with 36 regional referral centers (type A hospital), 52 provincial referral centers (type S), and 34 general hospitals (type M1). Ten out of a total of 13 university hospitals are also considered as having the capacity of the provincial or regional referral centers. Since Thailand achieved UHC in 2002, public hospital emergency departments (EDs) play a dominant role in the provision of emergency medical care for an annual estimated 13 million ED visitors nationwide. These public hospitals serve referral needs of patients with emergency conditions mostly from almost 800 district hospitals throughout the country. All patients visiting the Thai EDs were triaged using the 5 level Emergency Severity Index (ESI). In this system, level 5 indicates the lowest acuity (no resources needed to move the patient to final disposition from the ED), and level 1 indicates the highest acuity (intubated, apnoeic, pulseless or unresponsive). Inter-rater reliability between registered-nurse ESI level and the true ESI level was $\kappa = 0.89$; Pearson $r = 0.83$ ($P < 0.001$).[10]

2.2 | Study population

The study population included all regional referral centers (type A hospital), provincial referral centers (type S), general hospitals (type M1), and university hospitals that provide 24-hour emergency medical care. After the confirmation of the availability of ED doctors and nurses 24 hours through telephone calls, 130 ED sites were identified across Thailand as meeting the inclusion criteria (Table 1).

The data of surgical and inpatient capacity of targeted hospitals under MOPH (Table 2) were retrieved from the Geographical Information System for Health Resource website (http://www. http://gishealth.moph.go.th/healthmap/resource.php/, accessed 10 May 2018). Only reported data (n = 119) can be accessed.

2.3 | Survey tool development

The purpose of the survey was to obtain descriptive information on Thai EDs and their operational characteristics and governance functions. The mail survey was developed from conducting relevant literature review by two investigators (P.S. and S.S.) and in-depth interview with eight medical chief officers and six nurse chief officers from randomly selected EDs of three university hospitals and five MOPH service hospitals. It contained four parts: Part A related to workforce configuration, part B related to ED workload and operational characteristics, part C collected data on ED services for high-priority health burdens, and part D collected information related to governance functions. The final version was pilot tested with the participants in the above hospitals to ensure its national validity.

TABLE 1 Comparing percentage distribution by types of hospitals in hospital samples and target population of hospitals

Types of Hospitals	Target (%)	Response (n = 91)	Response Rate, %
M1	34 (26.2%)	24 (26.4%)	70.6
S	52 (40.0%)	38 (41.8%)	73.1
A	36 (27.7%)	23 (25.3%)	63.9
University	8 (6.2%)	6 (6.6%)	75.0
All	130 (100%)	91 (100%)	70.0

Twelve medical schools without hospitals were excluded from the target population.

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TABLE 2 Surgical and inpatient capacity of targeted service hospitals under Ministry of Public Health

Types of Hospitals	N	Total Beds	Average Beds per Hospital	Total ICU Beds	Average ICU Beds per Hospital	Total Operating Rooms	Average Operating Rooms per Hospital
M1	36	8205	227.9	431	12.0	169	4.7
S	49	18 534	378.2	1411	28.8	397	8.1
A	34	24 919	732.9	2251	66.2	518	15.2
All	119	51 658		4093		1084	

Abbreviation: ICU, intensive care unit.

Source: http://gishealth.moph.go.th/healthmap/resource.php.

2.4 | Data collection

The instrument was administered through a mail questionnaire survey to all ED chief nursing officers and ED chief medical officers in the target hospitals. To enhance the response rate, a follow-up call was scheduled at 2 weeks after sending the mail and follow-up letter after no response. A polo shirt was offered as survey incentives.

The respondents were advised to collect information from work shift schedules for part A, from electronic ED dataset for part B, and from hospital administrators for part D. Given routine service operations for part C, the information could be readily filled out by the respondents.

2.5 | Data analysis

Nonparametric continuous data were described as median and interquartile range (IQR), while categorical data were shown as the number and its percentage. The Pearson chi-squared test or Fisher exact test was used to compare categorical data where appropriate. The Kruskal-Wallis test was used to test the differences among nonparametric continuous data. Differences between groups were considered statistically significant if $P < 0.05$ (two-tailed). All statistics were analyzed using STATA version 15.2 software 2 (StataCorp. 2017. Stata Statistical Software: Release 15. College Station, TX: StataCorp LLC). More than 99% of questionnaires were complete across all sections.

3 | RESULTS

Surveys were completed for 91 of the 130 eligible Thai EDs, achieving a response rate of 70%. Twenty-one respondents (23%) were ED chief medical officers, and the rest was ED chief nursing officers. Percentage distribution of hospital types among responded hospitals was comparable with that of the target hospitals (Table 1). Table 2 depicted surgical and inpatient capacity of targeted hospitals under MOPH. Available figures (from online search) for each of 10 targeted university hospitals was the number of beds varying from 471 to 2300 (mean = 1309.8).

The median number of patients visiting hospital EDs was 51 221 per year with 32.8% considered nonurgent according to the ESI triage system (ESI levels 4-5) (Table 3). Nonetheless, M1 hospitals received the highest percentage of nonurgent patients (52.4%).

With regard to hospital ED workforce, Table 4 depicted types of full-time ER professionals in terms of median and IQR. University hospital EDs were staffed with a substantially higher number of emergency physician (EP) and EM residents than EDs of service-based secondary care and tertiary care hospitals under MOPH. Similarly, the

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TABLE 3 Median and IQR of ED visits per year

Types of hospitals	ED Visits		ESI					Nonurgent (ESI 4-5)
			1	2	3	4	5	
	Median	IQR	%	%	%	%	%	%
All	51 221	42 026, 60 458	3.4	9.9	38.7	23.2	9.6	32.8
M1	47 143	34 393, 58 150	1.9	7.7	30.6	25.2	27.2	52.4
S	48 248	38 475, 58 179	3.9	9	39.1	23.4	9.2	32.6
A	62 444	53 094, 72 134	6.4	16.1	46.6	20.1	5	25.1
University	46 827	40 581, 55 000	1.4	16.3	40.6	27.2	6.8	34

Abbreviations: ED, emergency department; IQR, interquartile range.

TABLE 4 Median and IQR of full-time personnel per shift by types of hospitals

Types of Personnels	Types of Hospitals	Median	IQR	*P* Value
Emergency physicians	M1	1	0, 1	<0.001
	S	1	0, 2	
	A	4	2, 5	
	University	8	7, 10	
EM residents	M1	0	0, 0	<0.001
	S	0	0, 0	
	A	0	0, 0	
	University	19.5	18, 22	
Interns	M1	6	1, 11	<0.001
	S	2	1, 10	
	A	0.5	0, 2	
	University	3	0, 10	
RNs	M1	15	12, 18	<0.001
	S	17	14, 22	
	A	29.5	24.5, 34	
	University	33	22, 44	
Emergency nurses	M1	2	1, 4	0.07
	S	2	1, 4	
	A	4	2.5, 6	
	University	4	2, 5	
EMTs	M1	2	1, 4	<0.001
	S	4	2, 6	
	A	7.5	3, 10	
	University	0	0, 0	
Paramedics	M1	0	0, 1	0.474
	S	0	0, 2	
	A	0	0, 3.5	
	University	0	0, 0	

Abbreviations: EM, emergency medicine; IQR, interquartile range; RN, registered nurse.

number of registered nurses (RNs) was found to be the highest for university hospital EDs. In the opposite, university hospital EDs did not report the presence of EMT. Regardless of the reported number of hospital ED workforce, almost all hospitals expressed concerns about the deficiency of doctors and nurses especially EP and emergency nurses (Table 5).

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TABLE 5 Number (%) of hospitals reporting insufficient personnel and using interns as main source of doctor services in ED

	Types of Hospitals										
	All	N = 91	M1	N = 25	S	N = 36	A	N = 24	University	N = 6	*P*
Types of Personnels	n	%	n	%	n	%	n	%	n	%	Value
All doctors	78	85.7	19	76.0	34	94.4	21	87.5	4	66.7	0.079
Emergency physicians	83	91.2	23	92.0	33	91.7	22	91.7	5	83.3	0.838
Interns as main doctors in ED	75	82.4	24	96.0	32	88.9	18	75.0	1	16.7	<0.001
All nurses	85	93.4	21	84.0	36	100.0	22	91.7	6	100.0	0.068
Emergency nurses	91	100.0	25	100.0	36	100.0	24	100.0	6	100.0	N/A

Abbreviation: ED, emergency department.

Given the workforce status and the workload, almost all hospitals reported the availability of specific services for emergency patients with acute myocardial infarction (MI), acute stroke, or trauma (Table 6). However, the percentage of hospitals reporting the availability of coronary artery catheterization (34.1%) and thrombolytic infusion for acute thrombotic stroke (24.2%) was limited.

Finally, the survey explored hospital perception of governance functions as obstacles to manage emergency systems. Almost all hospitals implicated three governance functions contributing to difficulties in addressing workforce shortage: public sector downsizing, the imbalance between ED staff quota and workload, and government financial regulation limiting hiring capacity of part-time staff (Table 7). In addition, government financial regulation was associated with difficulties for hospitals to innovate a new service model such as midnight urgent care.

TABLE 6 Number (%) of hospitals reporting availability of services for patients with acute MI, acute stroke, and trauma

	Types of Hospitals										
	All	N = 91	M1	N = 25	S	N = 36	A	N = 24	University	N = 6	*P*
Services	n	%	n	%	n	%	n	%	n	%	Value
MI											
MI fast track protocol	90	98.9	24	96.0	36	100.0	24	100.0	6	100.0	0.604
Thrombolytic transfusion	88	96.7	23	92.0	35	97.2	24	100.0	6	100.0	0.568
Coronary artery catheterization	31	34.1	1	4.0	3	8.3	21	87.5	6	100.0	<0.001
Stroke											
Stroke fast track protocol	90	98.9	24	96.0	36	100.0	24	100.0	6	100.0	0.604
Thrombolytic transfusion	22	24.2	4	16.0	7	19.4	7	29.2	4	66.7	0.069
Trauma											
24-h ultrasonography	88	96.7	24	96.0	36	100.0	22	91.7	6	100.0	0.279
24-h CT scan	89	97.8	23	92.0	36	100.0	24	100.0	6	100.0	0.269
24-h surgery	75	82.4	17	68.0	31	86.1	22	91.7	5	83.3	0.153

Abbreviations: CT, computed tomography; MI, myocardial infarction.

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TABLE 7 Number (%) of hospitals considering macropolicies compromising ED management

	All	N = 91	M1	N = 25	S	N = 36	A	N = 24	University	N = 6	*P*
Problems	n	%	n	%	n	%	n	%	n	%	Value
Workforce											
Public sector downsizing rendering personnel shortage	87	95.6	25	100.0	34	94.4	23	95.8	5	83.3	0.318
Imbalance between ED staff quota and workload	89	97.8	25	100.0	36	100	23	95.8	5	83.3	0.039
Government financial regulation limiting hiring capacity of part-time staffs	83	91.2	24	96.0	32	88.9	23	95.8	4	66.7	0.124
Service innovations											
Government financial regulation hampering service innovations	70	76.9	21	84.0	27	75	19	79.2	3	50.0	0.339

Abbreviation: ED, emergency department.

4 | DISCUSSION

4.1 | Workforce status vis-à-vis the workload

This study provides rare evidence at national scale profiling hospital emergency system in Thailand, a middle-income country with UHC. ED size in our study (50 000 visits per year; Table 2) is roughly similar to that of a national survey in Australia, 36.3% with less than 30 000 ED visits per year.[11]

To a certain extent, ED staffing seems to positively correlate with workloads (Tables 3 and 4). This is similar to findings from a study in Queensland (Australia) with ED nursing staff (nursing per 1000 weighted activity units) correlated with total ED presentations.[12] Exceptional status of university hospitals in our study was observed with fewer ED visits than the rest but a higher number of EPs, EM residents, and RNs. This may suggest the influence of teaching status of university hospitals on the allocation of the human resource.

In addition, our study dealt with some common agendas facing hospital EDs in developed and developing countries. On the other hand, our study reveals certain features of the Thai systems in contrast to those in LMICs.

Our study (Table 3) found that the overall figure (33%) for ED visits with nonurgent presentation (NUP) (ESI 4-5) was similar to the percentage (37% [range 8%-62%]) of all ED visits in the United States that were judged to be NUP.[13] The figures from a few LMICs such as Saudi Arabia and Iran were 20% to 72%.[14,15] However, these figures were based on varying definitions across studies, hence rendering difficulty in comparison.

Visiting the ED for NUP has raised two different concerns. The first deals with a concern about increasing cost of care, overcrowding, excessive use of investigations, and poor continuity of care.[16] The second concerns patient safety as evidence demonstrated that patients presenting to EDs have worse risk-adjusted short-term outcomes than those presenting to outpatient clinics, even after controlling for acuity level of visit or discharge diagnosis.[17]

4.2 | Governance obstacles

Suggested strategies to reduce NUP include financial disincentives such as higher co-payments, patient education, redirection of patients to an alternative source of care, and provision of after-hours and weekend care by primary care physician.[16] According to a cohort study of data from a large commercial health plan in the United States from

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2008 to 2015, ED visits per enrollee for the treatment of low-acuity conditions decreased by 36%, whereas utilization of non-ED acute care venues increased by 140%. These venues were urgent care centers, retail clinics, and telemedicine.[18] Implementation of these strategies may require flexibility for hospitals to set financial disincentives or mobilization and management of human resources. In the context of governance functions in Thailand, this may not be the case for the majority of public hospitals in our study given the ED management obstacles shown in Table 7.

4.3 | Availability of services for selected high-priority health burdens

Nonetheless, the hierarchical arrangement of Thai hospitals seems to reasonably influence the orderly percentage of ED visits with NUP across the hierarchy as shown in Table 3. This arrangement was also reflected in staff mix with university hospitals having substantially more number of EPs and EM residents than those of service hospitals (Table 4).

The hierarchy does not only determine differences in structure and staff mix (Tables 2 and 4) among the hospitals but may also include referral relationships as indicated by orderly sophistication among hospitals in the service provision for patients with acute MI, acute thrombotic stroke, or trauma (Table 6). In contrast to the scarcity of these advanced services in most LMICs,[19-21] nationwide availability of these services in Thai public hospitals is probably a result of the implementation of UHC since 2002 and the affordability of Thai health care systems (public expenditure on health was 12.7% of the annual government budget or less than 4% of gross domestic product [GDP][22]).

4.4 | Lessons learnt

The findings of the present study reveal that staff mix varies across hospital EDs reflecting the hierarchical arrangement of Thai health care systems. When examining operational models, similar patterns emerged in the availability of advanced care for patients with acute MI, acute thrombotic stroke, or trauma (Table 6). Nationwide availability of advanced care was made possible due to the implementation of UHC and affordability of Thai health care systems. Nonetheless, a better response to patients with NUP requires improved governance functions of the centralized health care system. The present study will enable us to examine associations between these service characteristics and patient outcomes of EDs.

5 | LIMITATIONS

The limitations of this study relate to the completeness of data collection. With a 70% response rate and an almost identical percentage of hospital types between our study and the target population (Table 1), the representativeness of our study samples is deemed considerably high. Our study was restricted to public EDs with the exclusion of private EDs, which represent almost 20% of all Thai EDs, and so limits the generalizability of the findings of the present study. In addition, and common to all survey studies, self-reported data are subject to threats of validity.

6 | CONCLUSIONS

The present study is the first work that measures the staffing configuration, organizational characteristics, and governance obstacles of Thai EDs. Through the present study, we know the service patterns and distribution of the staff mix. Under UHC and hierarchical arrangement of public hospital EDs, advanced care for patients with high-priority health burdens was available nationwide. Facing governance obstacles, the public hospital EDs probably have difficulty to apply evidence-based knowledge to cope with patients with NUP.

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DATA AVAILABILITY

All the data and questionnaire are available upon request to the corresponding author.

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REFERENCES

1. World Health Organization. *Sixtieth World Health Assembly. Resolution WHA 60.22: Emergency-Care Systems*. Geneva: World Health Organization; 2007 https://apps.who.int/iris/bitstream/handle/10665/22596/A60_R22-en.pdf?sequence=1&isAllowed=y.
2. Shanahan T, Risko N, Razzak J, Bhutta Z. Aligning emergency care with global health priorities. *Int J Emerg Med*. 2018;11(1):52. https://doi.org/10.1186/s12245-018-0213-8
3. World Health Organization. WHO emergency care system framework infographic. http://www.who.int/emergencycare/emergencycare_infographic/en/
4. Kruk ME, Gage AD, Joseph NT, Danaei G, García-Saisó S, Salomon JA. Mortality due to low-quality health systems in the universal health coverage era: a systematic analysis of amenable deaths in 137 countries. *Lancet (London, England)*. 2018;392(10160):2203-2212. https://doi.org/10.1016/S0140-6736(18)31668-4
5. Japiong KB, Asiamah G, Owusu-Dabo E, et al. Availability of resources for emergency care at a second-level hospital in Ghana: a mixed methods assessment. *Afr J Emerg Med: Rev Africaine de La Medecine d'urgence*. 2016;6(1):30-37. https://doi.org/10.1016/j.afjem.2015.06.006
6. Coyle RM, Harrison H-L. Emergency care capacity in Freetown, Sierra Leone: a service evaluation. *BMC Emerg Med*. 2015;15(1):2. https://doi.org/10.1186/s12873-015-0027-4
7. Hammerstedt H, Maling S, Kasyaba R, et al. Addressing World Health Assembly Resolution 60.22: a pilot project to create access to acute care services in Uganda. *Ann Emerg Med*. 2014;64(5):461-468. https://doi.org/10.1016/j.annemergmed.2014.01.035
8. Aluisio AR, Barry MA, Martin KD, et al. Impact of emergency medicine training implementation on mortality outcomes in Kigali, Rwanda: an interrupted time-series study. *Afr J Emerg Med*. 2018;9(1):14-20. https://doi.org/10.1016/j.afjem.2018.10.002
9. Tangcharoensathien V, Witthayapipopsakul W, Panichkriangkrai W, Patcharanarumol W, Mills A. Health systems development in Thailand: a solid platform for successful implementation of universal health coverage. *Lancet*. 2018;391(10126):1205-1223. https://doi.org/10.1016/S0140-6736(18)30198-3
10. Tanabe P, Gimbel R, Yarnold PR, Kyriacou DN, Adams JG. Reliability and Validity of Scores on the Emergency Severity Index Version 3. *Acad Emerg Med*. 2004;11(1):59-65. https://doi.org/10.1197/j.aem.2003.06.013
11. Gardner G, Gardner A, Middleton S, et al. Mapping workforce configuration and operational models in Australian emergency departments: a national survey. *Aust Health Rev*. 2018;42(3):340-347. Retrieved from https://doi.org/10.1071/AH16231
12. Toloo G-S, Burke J, Crilly J, et al. Understanding ED performance after the implementation of activity-based funding. *Int J Health Plann Manag*. 2018;33(2):405-413. https://doi.org/10.1002/hpm.2475
13. Uscher-Pines L, Pines J, Kellermann A, Gillen E, Mehrotra A. Emergency department visits for nonurgent conditions: systematic literature review. *Am J Manag Care*. 2013;19(1):47-59. Retrieved from https://www.ncbi.nlm.nih.gov/pubmed/23379744
14. Dawoud SO, Ahmad AMK, Alsharqi OZ, Al-Raddadi RM. Utilization of the emergency department and predicting factors associated with its use at the Saudi Ministry of Health General Hospitals. *Global J Health Sci*. 2015;8(1):90-106. https://doi.org/10.5539/gjhs.v8n1p90

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15. Jalili M, Shirani F, Hosseininejad M, Asl-E-Soleimani H. Emergency department nonurgent visits in Iran: prevalence and associated factors. *Am J Manag Care*. 2013;19(1):e1-e8.
16. Alele FO, Emeto TI, Callander EJ, Watt K. Non-urgent paediatric emergency department presentation: a systematic review. *J Paediatr Child Health*. 2018;0(0). https://doi.org/10.1111/jpc.14352
17. Niedzwiecki M, Baicker K, Wilson M, Cutler DM, Obermeyer Z. Short-term outcomes for Medicare beneficiaries after low-acuity visits to emergency departments and clinics. *Med Care*. 2016;54(5):498-503. https://doi.org/10.1097/MLR.0000000000000513
18. Poon SJ, Schuur JD, Mehrotra A. Trends in visits to acute care venues for treatment of low-acuity conditions in the United States from 2008 to 2015. *JAMA Intern Med*. 2018;178(10):1342-1349. https://doi.org/10.1001/jamainternmed.2018.3205
19. Yan LL, Li C, Chen J, et al. Prevention, management, and rehabilitation of stroke in low- and middle-income countries. *eNeurologicalSci*. 2016;2:21-30. https://doi.org/10.1016/j.ensci.2016.02.011
20. Vedanthan R, Seligman B, Fuster V. Global perspective on acute coronary syndrome: a burden on the young and poor. *Circ Res*. 2014;114(12):1959-1975. https://doi.org/10.1161/CIRCRESAHA.114.302782
21. Kiragu AW, Dunlop SJ, Mwarumba N, et al. Pediatric trauma care in low resource settings: challenges, opportunities, and solutions. *Front Pediatr*. 2018;6:155. https://doi.org/10.3389/fped.2018.00155
22. Tangcharoensathien V, Pitayarangsarit S, Patcharanarumol W, et al. Promoting universal financial protection: how the Thai universal coverage scheme was designed to ensure equity. *Health Res Pol Syst*. 2013;11(1):25. https://doi.org/10.1186/1478-4505-11-25.

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